Conditionally Whitened Generative Models For Probabilistic Time Series Forecasting

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2025/10/09

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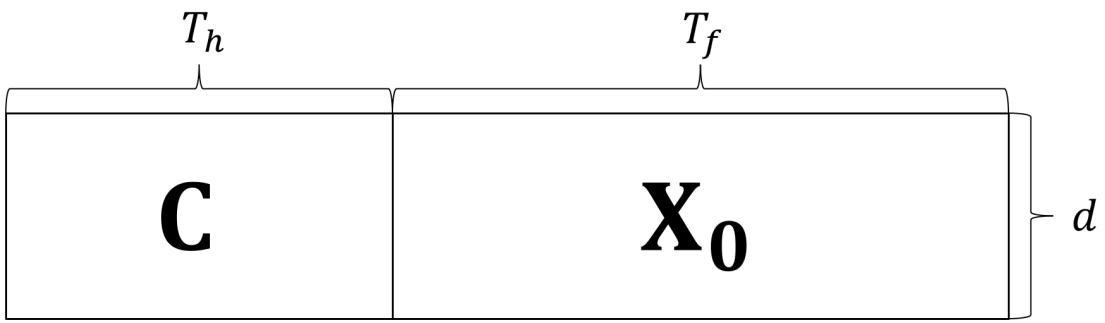
1. Probabilistic Time Series Forecasting: Task & Challenges

Some Definitions

- d: Dimensionality of the time series
- T_h: Historical sequence length (fixed in experiments)
- T_f: Forecast horizon (long-term forecasting setting)
- $\mathbf{C}$: historical observations in $\mathbb{R}^{d \times T_h}$
- $\mathbf{X}_0$: future time series in $\mathbb{R}^{d \times T_f}$

The task of probabilistic time series forecasting (TSF) is to learn the distribution of future time series $\mathbf{X}_0$ given the historical observations $\mathbf{C}$. We can denote the conditional distribution of $\mathbf{X}_0|\mathbf{C}$ as $P_{\mathbf{X}|\mathbf{C}}$.



Challenges

TSF and probabilistic TSF have been extremely important for decision-making in various domains such as finance, healthcare, environmental science and transportation. But there are several challenges:

1. **Non-stationarity**: Long-term trends, seasonality, and heteroscedasticity
2. **Complex inter-variable dependencies**: Intricate correlations between variables
3. **Inherent data uncertainty**: Short-term fluctuations and noise
4. **Distribution shift**: Distribution shift between training and test data

2. Recent Generative Models for TSF

Diffusion Models for Probabilistic TSF

Diffusion models have become mainstream for probabilistic time series forecasting, with representative models listed below.

Model	Short description
TimeGrad (2021, Rasul et al.)	Autoregressive model, use RNN to encode **C**
CSDI (2021, Tashiro et al.)	Non-autoregressive model, use Transformer as the backbone
SSSD (2023, Alcaraz & Strodthoff)	Non-autoregressive model, use structured state space model as the backbone
Diffusion-TS (2024, Yuan & Qiao)	Non-autoregressive model, the backbone is able to decompose the trend/seasonal/residual
Flow-TS (2025, Hu & Wang)	Same backbone with Diffusion-TS, use flow matching

Limitations

These models are vulnerable to non-stationary components of time series, such as trends, and heteroscedasticity. And they are not robust to the distribution shift between training/test datasets.

Prior-Informed Models

Several previous models have shown that prior models derived from $\mathbf{C}$ can substantially improve the quality of diffusion models.

Model	Core Design
TMDM (2024, Li et al.)	Estimate $\mathbb{E}[\mathbf{X}_0 \mid \mathbf{C}]$, and incorporate it into the diffusion processes
S2DBM	Same with TMDM, but use diffusion bridge model
NsDiff (2025, Ye et al.)	Estimate $\mathbb{E}[\mathbf{X}_0 \mid \mathbf{C}]$ and sliding-window variance of $\mathbf{X}_0$, and incorporate them into diffusion processes. It is the SOTA model currently

Limitations

- NsDiff ignores inter-variable covariance (the sliding-window variance is diagonal);
- The reverse process of NsDiff is *overly complex* – requires solving d univariate quadratic equations, leading to potential numerical failures;
- *Inconsistent terminal distribution* — the forward process converges to a Gaussian distribution with the true sliding-window variance of $\mathbf{X}_0$, but the reverse process starts from a Gaussian distribution with the estimated sliding window variance.

A clarification

The term 'prior' here refers to an auxiliary model that estimates the conditional mean, conditional variance, or other numerical characteristics. It does not refer to other meanings of 'prior', such as physical constraints or similar concepts.

Motivation

The previous works show that using prior models can improve the generative models. But some key questions about prior models are unanswered:

1. How do priors models improve performance and what accurate should the prior models be?
2. How to train such priors models effectively?
3. Are existing prior-integration mechanisms redundant, and can they be simplified in diffusion frameworks?

3. Theoretical Foundation

How do priors models improve performance?

The total variation (TV) error bound of a diffusion model and the true distribution is effected by the **convergence error of the forward process**, see (2023, Oko; 2023, Chen; 2024, Fu).

The convergence error involves the **Kullback–Leibler divergence (KLD) between the true conditional distribution and the terminal distribution of the diffusion model**.

In other words, if we replace the terminal distribution of the diffusion model with one that is closer to the true data distribution, the generated samples will match the true data distribution more closely.

Notations for theorem

Let $X_0 \in \mathbb{R}^{d_x}$ and $C \in \mathbb{R}^{d_c}$ be two general random variables, and the conditional distribution of X_0 given C is denoted as $P_{X_0|C}$. The true conditional mean and conditional covariance of X_0 given C are denoted as $\mu_{X|C}$ and $\Sigma_{X|C}$. Let $\widehat{\mu}_{X|C}$ and $\widehat{\Sigma}_{X|C}$ be the estimators of $\mu_{X|C}$ and $\Sigma_{X|C}$.

Take DDPM as an example, the TV error bound is effected by $D_{\mathrm{KL}}\big(P_{X|C} \,\|\, N(0, I_{d_x})\big)$, where $N(0, I_{d_x})$ is the terminal distribution of DDPM.

We can replace the terminal distribution with $N(\widehat{\mu}_{X|C}, \widehat{\Sigma}_{X|C})$. Then, the KLD becomes $D_{\mathrm{KL}}\Big(P_{X|C} \,\|\, N(\widehat{\mu}_{X|C}, \widehat{\Sigma}_{X|C})\Big)$. So, the advantage of this replacement can then be measured by the reduction in

$$D_{\mathrm{KL}}\Big(P_{X|C} \,\|\, N(\widehat{\mu}_{X|C}, \widehat{\Sigma}_{X|C})\Big) \quad \text{relative to} \quad D_{\mathrm{KL}}\big(P_{X|C} \,\|\, N(0, I_{d_x})\big)\,.$$

Theorem 1

Let $P_{X|C}$ is true conditional distribution of $X \in \mathbb{R}^{d_x}$ given C; denote $Q_0 = N(0, I_{d_x})$, $\widehat{Q} = N(\widehat{\mu}_{X|C}, \widehat{\Sigma}_{X|C})$; and let $\widehat{\lambda}_{X|C,i}$ be the i-th eigenvalue of $\widehat{\Sigma}_{X|C}$. The sufficient condition of $D_{\mathrm{KL}}\Big(P_{X|C} \,\|\, N(\widehat{\mu}_{X|C}, \widehat{\Sigma}_{X|C})\Big) \leq D_{\mathrm{KL}}\big(P_{X|C} \,\|\, N(0, I_{d_x})\big)$ is:

$$\begin{aligned}\left(\min_{i\in\{1,\ldots,d_x\}} \widehat{\lambda}_{X|C,i}\right)^{-1} \left(\|\mu_{X|C} - \widehat{\mu}_{X|C}\|_2^2 + \|\Sigma_{X|C} - \widehat{\Sigma}_{X|C}\|_N\right) \\ + \sqrt{d_x}\,\|\Sigma_{X|C} - \widehat{\Sigma}_{X|C}\|_F \leq \|\mu_{X|C}\|_2^2,\end{aligned} \tag{1}$$

where $\|\Sigma_{X|C} - \widehat{\Sigma}_{X|C}\|_N = \sum_{i=1}^{d_x} \widetilde{s}_i$ ($\widetilde{s}_i$ is the i-th singular value of $\Sigma_{X|C} - \widehat{\Sigma}_{X|C}$).

Some comments on Theorem 1

1. **Theoretical guarantee**: the sufficient condition of reducing the KLD.
2. **Factors influencing the sufficient condition**: the accuracy of $\widehat{\mu}_{X|C}$ and $\widehat{\Sigma}_{X|C}$, and the scale of the minimum eigenvalue of $\widehat{\Sigma}_{X|C}$, and the magnitude of $\mu_{X|C}$.
3. **The basis of loss function**: tell us how to design the loss functions for learning $\mu_{X|C}$ and $\Sigma_{X|C}$.

4. Joint Mean and Covariance Estimator (JMCE)

Joint Mean and Covariance Estimator (JMCE)

We first estimate the conditional mean and covariance jointly. The true covariance of time series is extremely hard to learn, so we chose to learn the sliding-window covariance $\widetilde{\Sigma}_{\mathbf{X}_0,t}, t = 1, \ldots, T_f$, which is an alternative of the true covariance.

Let a Non-stationary Transformer (2022, Liu) be the backbone of our JMCE, outputing $\widehat{\mu}_{\mathbf{X}|\mathbf{C}}, \widehat{L}_{1|\mathbf{C}}, \ldots, \widehat{L}_{T_f|\mathbf{C}}$, where $\widehat{\mu}_{\mathbf{X}|\mathbf{C}}$ is the estimator of conditional mean, and $\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}} := \widehat{L}_{t|\mathbf{C}}\widehat{L}_{t|\mathbf{C}}^\top$ is the estimators of $\widetilde{\Sigma}_{\mathbf{X}_0,t}$, for $t = 1, \ldots, T_f$. We aim to design a loss function to minimize $\mathcal{L}_2 := \mathbb{E}\left\|\mathbf{X}_0 - \widehat{\mu}_{\mathbf{X}|\mathbf{C}}\right\|_2^2$, $\mathcal{L}_F := \mathbb{E}\sum_{t=1}^{T_f}\left\|\widetilde{\Sigma}_{\mathbf{X}_0,t} - \widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}}\right\|_F$ and $\mathcal{L}_{\text{SVD}} := \mathbb{E}\sum_{t=1}^{T_f}\left\|\widetilde{\Sigma}_{\mathbf{X}_0,t} - \widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}}\right\|_N$.

The penalty of minimum eigenvalue

In (1), the sufficient condition is also influenced by the minimum eigenvalues of $\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}}, t = 1, \ldots, T_f$. Thus, we design a novel penalty term to force the minimum eigenvalue to stay from 0:

$$\mathcal{R}_{\lambda_{\min}}\left(\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}}\right) := \sum_{i=1}^{d} \mathrm{ReLU}\left(\lambda_{\min} - \widehat{\lambda}_{\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}},i}\right), \tag{2}$$

where $\lambda_{\min}$ is a hyperparameter and $\widehat{\lambda}_{\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}},i}(i = 1, \ldots, d)$ denote the eigenvalues of $\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}}$. Any eigenvalue smaller than $\lambda_{\min}$ will be penalized.

Finally, the loss function of JMCE is:

$$\mathcal{L}_{\mathrm{JMCE}} = \mathcal{L}_2 + \mathcal{L}_{\mathrm{SVD}} + \lambda_{\min}\sqrt{d \cdot T_f}\mathcal{L}_F + w_{\mathrm{Eigen}} \cdot \sum_{t=1}^{T_f} \mathcal{R}_{\lambda_{\min}}\left(\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}}\right), \tag{3}$$

where w_{Eigen} is a hyperparameter that controls the strength of the penalty.

5. Condtionally Whitened Diffusion Models (CW-Diff) and Condtionally Whitened Flow Matching (CW-Flow)

Define a tensor operation

For simplicity, we define $\widehat{\Sigma}^k_{\mathbf{X}_0|\mathbf{C}} := [\widehat{\Sigma}^k_{\mathbf{X}_0,1|\mathbf{C}}, \ldots, \widehat{\Sigma}^k_{\mathbf{X}_0,T_f|\mathbf{C}}] \in \mathbb{R}^{d\times d\times T_f}$ for $k \in \{-0.5, 0.5\}$ and $\boldsymbol{\epsilon} := [\epsilon_1, \ldots, \epsilon_{T_f}] \in \mathbb{R}^{d\times T_f}$.
We define the tensor operation

$$\widehat{\Sigma}^{0.5}_{\mathbf{X}_0|\mathbf{C}} \circ \boldsymbol{\epsilon} := [\widehat{\Sigma}^{0.5}_{\mathbf{X}_0,1|\mathbf{C}} \cdot \epsilon_1, \ldots, \widehat{\Sigma}^{0.5}_{\mathbf{X}_0,T_f|\mathbf{C}} \cdot \epsilon_{T_f}] \in \mathbb{R}^{d\times T_f}. \tag{4}$$

Accordingly, we say that a tensor follows $\mathcal{N}(\widehat{\mu}_{\mathbf{X}|\mathbf{C}}, \widehat{\Sigma}_{\mathbf{X}_0|\mathbf{C}})$ if it has the same distribution as $\widehat{\Sigma}^{0.5}_{\mathbf{X}_0|\mathbf{C}} \circ \boldsymbol{\epsilon} + \widehat{\mu}_{\mathbf{X}|\mathbf{C}}$.

Conditionally Whitened Diffusion (CW-Diff)

With the previous definition, we can use an OU process with drift and covariance to diffuse the original time series $\mathbf{X}_0$ to $\mathcal{N}(\widehat{\mu}_{\mathbf{X}|\mathbf{C}}, \widehat{\Sigma}_{\mathbf{X}_0|\mathbf{C}})$:

$$d\left(\mathbf{X}_\tau - \widehat{\mu}_{\mathbf{X}|\mathbf{C}}\right) = -\tfrac{1}{2}\beta_\tau\left(\mathbf{X}_\tau - \widehat{\mu}_{\mathbf{X}|\mathbf{C}}\right)d\tau + \sqrt{\beta_\tau}\cdot\widehat{\Sigma}^{0.5}_{\mathbf{X}_0|\mathbf{C}} \circ d\mathbf{W}_\tau,\ \mathbf{X}_0 \sim P_{\mathbf{X}|\mathbf{C}}, \quad (5)$$

for $\tau \in [0,1]$ and $\mathbf{W}_\tau$ is a Brownian motion in $\mathbb{R}^{d \times T_f}$.

Furthermore, the following SDE is equivalent to (5):

$$d\,\widehat{\Sigma}^{-0.5}_{\mathbf{X}_0|\mathbf{C}} \circ \left(\mathbf{X}_\tau - \widehat{\mu}_{\mathbf{X}|\mathbf{C}}\right) = -\tfrac{1}{2}\beta_\tau \cdot \widehat{\Sigma}^{-0.5}_{\mathbf{X}_0|\mathbf{C}} \circ \left(\mathbf{X}_\tau - \widehat{\mu}_{\mathbf{X}|\mathbf{C}}\right)d\tau + \sqrt{\beta_\tau}\,d\mathbf{W}_\tau,\ \tau \in [0,1],$$

which implies that the diffusion processes can be directly performed on $\mathbf{X}_0^{\mathrm{CW}} := \widehat{\Sigma}^{-0.5}_{\mathbf{X}_0|\mathbf{C}} \circ \left(\mathbf{X}_0 - \widehat{\mu}_{\mathbf{X}|\mathbf{C}}\right)$. We call this operation conditional whitening (CW).

Conditionally Whitened Diffusion (CW-Diff)

The CW operation subtracting $\widehat{\mu}_{\mathbf{X}|\mathbf{C}}$ and de-correlate every time point in $\mathbf{X}_0$ given $\mathbf{C}$, eliminating the non-stationary components and simplifying the inter correlation. And the forward process of CW-Diff has the same form with DDPM:

$$d\mathbf{X}_\tau^{\mathrm{CW}} = -\tfrac{1}{2}\beta_\tau \mathbf{X}_\tau^{\mathrm{CW}} d\tau + \sqrt{\beta_\tau} d\mathbf{W}_\tau,\ \tau \in [0,1], \tag{6}$$

which remains the diffusion processes simple. The loss function and the reverse process is same with DDPM. Assume the final sample generated by reverse process is $\overleftarrow{\mathbf{X}}_{\tau_{\min}}^{\mathrm{CW}}$, we just need to reverse the CW operation:

$$\overleftarrow{\mathbf{X}}_{\tau_{\min}} = \widehat{\Sigma}_{\mathbf{X}_0|\mathbf{C}}^{0.5} \circ \overleftarrow{\mathbf{X}}_{\tau_{\min}}^{\mathrm{CW}} + \widehat{\mu}_{\mathbf{X}|\mathbf{C}},$$

and we get the sample approximates $P_{\mathbf{X}|\mathbf{C}}$.

Algorithm of CW-Diff

Algorithm of training a CW-Diff:

1. Conditionally whiten $\mathbf{X}_0$ given $\mathbf{C}$: $\mathbf{X}_0^{\mathrm{CW}} = \widehat{\Sigma}_{\mathbf{X}_0|\mathbf{C}}^{-0.5} \circ (\mathbf{X}_0 - \widehat{\mu}_{\mathbf{X}|\mathbf{C}})$
2. Initialize a score network s_θ^{CW}
3. Minimize $\mathbb{E}\|s_\theta^{\mathrm{CW}}\left(\alpha_\tau \mathbf{X}_0^{\mathrm{CW}} + \sigma_\tau \boldsymbol{\epsilon}, \mathbf{C}, \tau\right) + \boldsymbol{\epsilon}/\sigma_\tau\|^2$, where $\alpha_\tau = \exp\left\{-\int_0^\tau \beta_s ds/2\right\}$ and $\sigma_\tau^2 = 1 - \alpha_\tau^2$

Algorithm of sampling by CW-Diff:

1. Draw $\overleftarrow{\mathbf{X}}_1^{\mathrm{CW}} \sim \mathcal{N}(0, I_{d\times d\times T_f})$
2. Solve SDE $d\overleftarrow{\mathbf{X}}_\tau^{\mathrm{CW}} = \left[-\frac{1}{2}\beta_\tau \overleftarrow{\mathbf{X}}_\tau^{\mathrm{CW}} - \beta_\tau s_\theta^{\mathrm{CW}}(\overleftarrow{\mathbf{X}}_\tau^{\mathrm{CW}}, \mathbf{C}, \tau)\right] d\tau + \sqrt{\beta_\tau} d\overleftarrow{\mathbf{W}}_\tau$ from $\tau = 1$ to $\tau = \tau_{\min}$, and get $\overleftarrow{\mathbf{X}}_{\tau_{\min}}^{\mathrm{CW}}$
3. Reverse the CW operation by $\overleftarrow{\mathbf{X}}_{\tau_{\min}} = \widehat{\Sigma}_{\mathbf{X}_0|\mathbf{C}}^{0.5} \circ \overleftarrow{\mathbf{X}}_{\tau_{\min}}^{\mathrm{CW}} + \widehat{\mu}_{\mathbf{X}|\mathbf{C}}$, where $\widehat{\mu}_{\mathbf{X}_0|\mathbf{C}}$ and $\widehat{\Sigma}_{\mathbf{X}_0|\mathbf{C}}^{0.5}$ are calculated by JMCE prior

Conditionally Whitened Flow Matching (CW-Flow)

In CW-Diff, the inverse matrices of $\widehat{\Sigma}_{\mathbf{X}_0,t|\mathbf{C}}$ need to be computed, with a complexity of $\mathcal{O}(d^3 T_f)$. To reduce this cost and improve efficiency, we transition to the Flow Matching framework.

Compared to the original flow matching, our Conditionally Whitened Flow Matching (CW-Flow) just changes the terminal distribution.

The Algorithm of CW-Flow

Algorithm of training CW-Flow:

1. Calculate $\widehat{\mu}_{\mathbf{X}_0|\mathbf{C}}$ and $\widehat{\Sigma}^{0.5}_{\mathbf{X}_0|\mathbf{C}}$ by JMCE prior
2. Initialize a vector field v^{CW}_{ψ}
3. Draw $\boldsymbol{\epsilon}^{\mathrm{CW}} \sim \mathcal{N}(0, I_{d\times d\times T_f})$ and calculate $\boldsymbol{\epsilon}^{\mathrm{CW}} = \widehat{\Sigma}^{0.5}_{\mathbf{X}_0|\mathbf{C}} \circ \boldsymbol{\epsilon}^{\mathrm{CW}} + \widehat{\mu}_{\mathbf{X}|\mathbf{C}}$
4. Minimize $\mathbb{E}\|\boldsymbol{\epsilon}^{\mathrm{CW}} - \mathbf{X}_0 - v^{\mathrm{CW}}_{\psi}(\mathbf{X}_0 + \tau(\boldsymbol{\epsilon}^{\mathrm{CW}} - \mathbf{X}_0), \mathbf{C}, \tau)\|^2$

1. Draw $\overleftarrow{\mathbf{X}}^{\mathrm{CW}}_1 \sim \mathcal{N}(0, I_{d\times d\times T_f})$ and calculate $\overleftarrow{\mathbf{X}}^{\mathrm{CW}}_1 = \widehat{\Sigma}^{0.5}_{\mathbf{X}_0|\mathbf{C}} \circ \overleftarrow{\mathbf{X}}^{\mathrm{CW}}_1 + \widehat{\mu}_{\mathbf{X}|\mathbf{C}}$
2. Solve ODE $d\overleftarrow{\mathbf{X}}^{\mathrm{CW}}_{\tau} = -v^{\mathrm{CW}}_{\psi}(\overleftarrow{\mathbf{X}}^{\mathrm{CW}}_{\tau}, \mathbf{C}, \tau)d\tau$ from $\tau = 1$ to $\tau = \tau_{\min}$, and get $\overleftarrow{\mathbf{X}}^{\mathrm{CW}}_{\tau_{\min}}$

The flow chart of JMCE, CW-Diff and CW-Flow

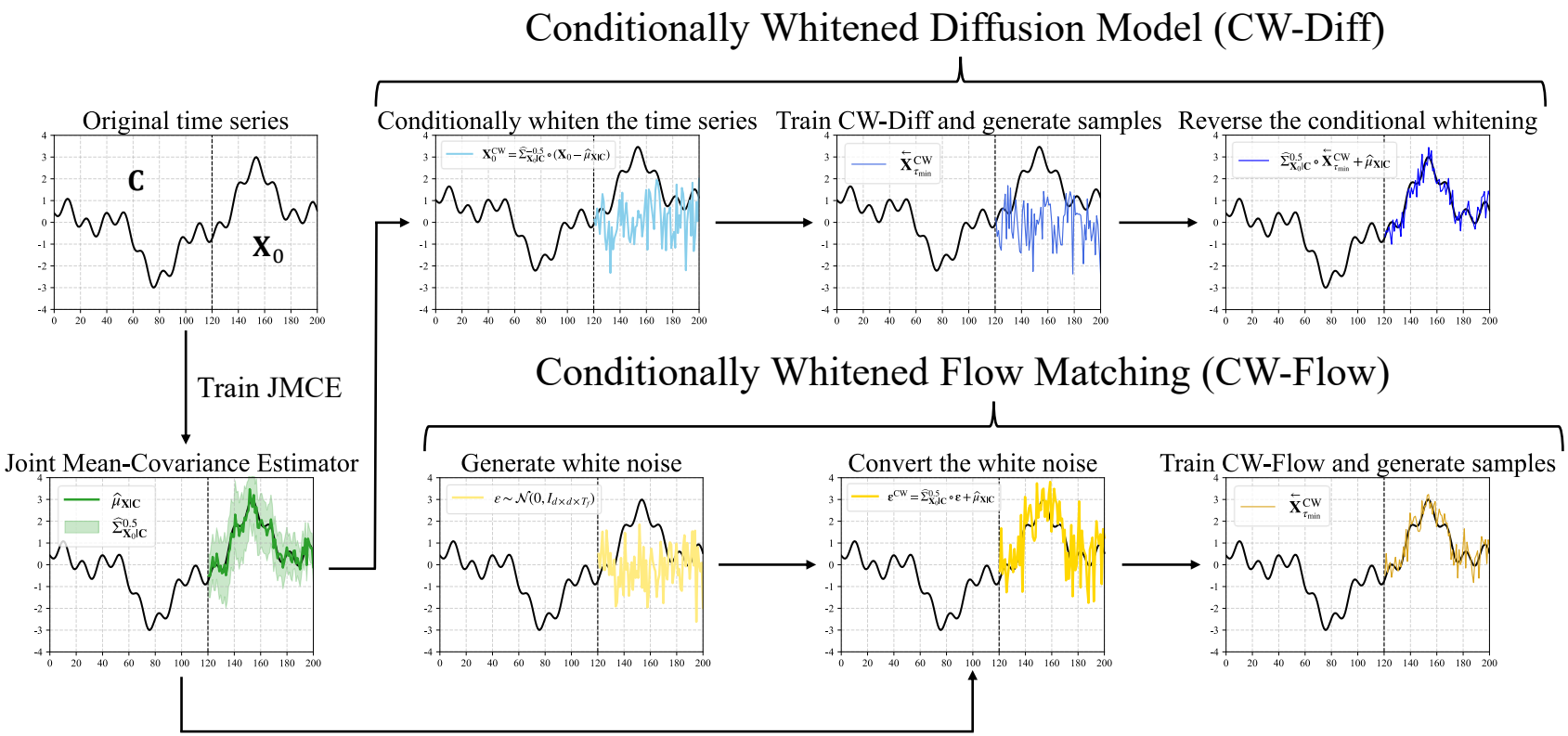


Figure: CW-Gen Framework: JMCE, CW-Diff and CW-Flow

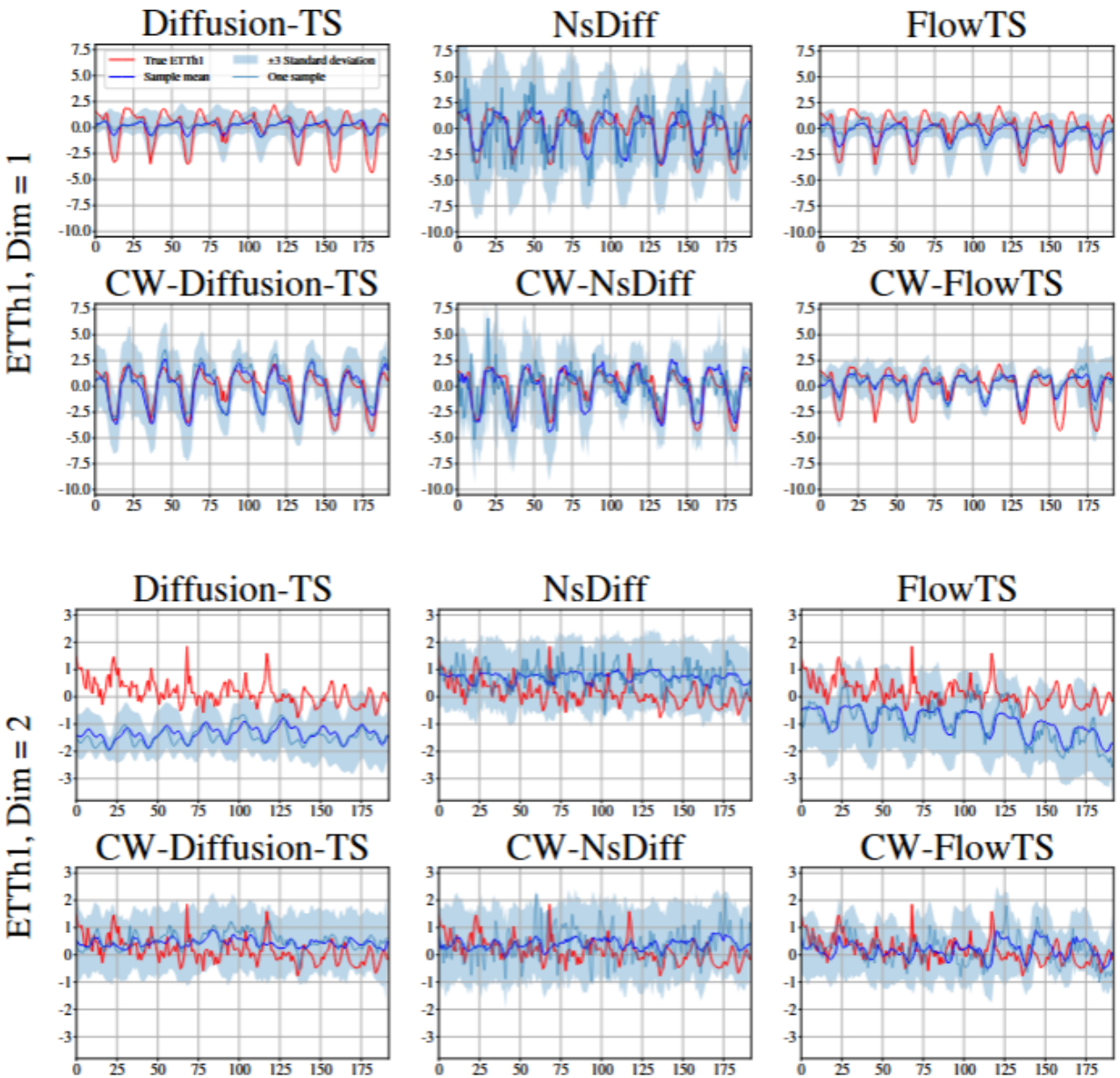
6. Experimental Results

Experimental Results

Model	CRPS↓		QICE↓		ProbCorr↓		Cond. FID↓	
	Raw	CW	Raw	CW	Raw	CW	Raw	CW
TimeDiff (2023)	0.736 (0.031)	0.530 (0.019)	9.513 (0.589)	12.610 (0.904)	0.299 (0.017)	0.238 (0.020)	14.081 (6.778)	3.998 (0.386)
SSSD (2023)	0.836 (0.153)	0.541 (0.064)	11.624 (1.312)	4.710 (1.555)	0.326 (0.032)	0.254 (0.030)	40.887 (17.601)	35.645 (19.248)
Diffusion-TS (2024)	0.641 (0.027)	0.453 (0.023)	6.742 (1.610)	2.856 (1.281)	0.337 (0.029)	0.261 (0.028)	21.098 (6.391)	18.432 (7.516)
TMDM (2024)	0.472 (0.031)	0.469 (0.027)	3.360 (1.055)	3.205 (0.731)	0.230 (0.014)	0.205 (0.009)	9.931 (4.439)	9.846 (4.037)
NSDiff (2025)	0.407 (0.032)	0.416 (0.015)	1.792 (0.682)	1.534 (0.314)	0.214 (0.014)	0.213 (0.008)	35.261 (7.785)	20.278 (5.912)
FlowTS (2025)	0.578 (0.065)	0.467 (0.014)	6.300 (1.329)	3.456 (0.607)	0.284 (0.024)	0.227 (0.011)	19.442 (13.874)	14.553 (9.103)
Win Rate	16.7%	**83.3%**	16.7%	**83.3%**	0.0%	**100.0%**	0.0%	**100.0%**

Table: Metrics for models trained on original ETTh1 (Raw) and conditionally whitened ETTh1(CW).

Experimental Results



Comparison of Diffusion-TS, NsDiff, FlowTS, and their CW variants on ETTh1.

In the first dimension, CW-Gen fits better.

In the second dimension, the distribution shift is mitigated.

Results on More Datasets

Table: Results on ETTh1, ETTh2, ILI, Weather, Solar Energy and win rates of our CW methods. Win rate refers to the proportion that our CW method outperforms original method on 6 models $\times$ 4 metrics.

Dataset	Win rate of CW-Gen
ETTh1	$22/24 \approx 91.67\%$
ETTh2	$22/24 \approx 91.67\%$
ILI	$21/24 \approx 87.50\%$
Weather	$22/24 \approx 91.67\%$
Solar Energy	$19/24 \approx 79.17\%$

7. Conclusion

Conclusion

1. **CW-Gen Framework**: Proposed a unified conditional generation framework **CW-Gen** with two instantiations—**CW-Diff** and **CW-Flow**.
2. **Theoretical Support**: Provided **sufficient conditions** to guarantee the improvement of CW-Gen.
3. **JMCE**: Designed Joint Mean–Covariance Estimator (**JMCE**) to estimate the conditional mean and sliding-window covariance jointly, and control the minimum eigenvalues simultaneously.
4. **Empirical Results**: Integrated CW-Gen with 6 SOTA generative models; tested on 5 real datasets. The results confirm better capture of non-stationarity, improved modeling of inter-variable dependencies, higher sample quality, and better mitigation of distribution shifts.